

Theory Assignment 2

GNR 638: Machine Learning for Remote Sensing II

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Problem Statement

Consider a linear autoencoder with:

- A input layer of dimension d .
- A hidden/"bottleneck" layer of dimension k (with $k < d$).
- A reconstruction layer also of dimension d .

Let the encoder be described by a weight matrix $\mathbf{W} \in \mathbb{R}^{k \times d}$ and bias $\mathbf{b} \in \mathbb{R}^k$, and the decoder by another weight matrix $\mathbf{V} \in \mathbb{R}^{d \times k}$ and bias $\mathbf{c} \in \mathbb{R}^d$. For a dataset $\{\mathbf{x}^{(n)}\}_{n=1}^N$ with $\mathbf{x}^{(n)} \in \mathbb{R}^d$, define the (mean-squared) reconstruction loss as:

$$\mathcal{L} = \sum_{n=1}^N \|\mathbf{x}^{(n)} - \hat{\mathbf{x}}^{(n)}\|^2,$$

where

$$\hat{\mathbf{x}}^{(n)} = \mathbf{V}(\mathbf{W}\mathbf{x}^{(n)} + \mathbf{b}) + \mathbf{c}$$

1. **Show** that, after removing any mean offset of the data via preprocessing (so you can assume $\sum_{n=1}^N \mathbf{x}^{(n)} = 0$ and drop the bias terms $\mathbf{b}, \mathbf{c}$ for simplicity), the weight matrices $\mathbf{W}$ and $\mathbf{V}$ that minimize $\mathcal{L}$ satisfy

$$\mathbf{V} = \mathbf{W}^T$$

2. **Prove** that the optimal matrix $\mathbf{W}^*$ (which also implies $\mathbf{V}^* = (\mathbf{W}^*)^T$) forms an orthonormal basis for the top k principal components of the data covariance matrix

$$\Sigma = \frac{1}{N} \sum_{n=1}^N \mathbf{x}^{(n)}(\mathbf{x}^{(n)})^T$$

In other words, the rows of $\mathbf{W}^*$ span the same subspace as the eigenvectors of Σ associated with its largest k eigenvalues.

Proof

As stated in the Problem statement, after removing any mean offset of the data via preprocessing, it can be assumed that $\sum_{n=1}^N x^{(n)} = 0$ and after dropping the bias terms for simplicity, the expression of reconstructed data becomes:

$$\hat{x}^n = V(Wx^{(n)} + b) + c = V(Wx^{(n)})$$

The mean-squared reconstruction loss is:

$$\mathcal{L} = \sum_{n=1}^N \|x^{(n)} - \hat{x}^{(n)}\|^2 = \sum_{n=1}^N \|x^{(n)} - VWx^{(n)}\|^2$$

Writing the above equation in matrix form ($X \in \mathbb{R}^{N \times d}$), where $\|\cdot\|_F$ is the Frobenius norm.

$$\mathcal{L} = \|X - XW^T V^T\|_F^2$$

Taking derivatives w.r.t. V :

$$\frac{\partial \mathcal{L}}{\partial V} = -2X^T XW^T + 2VW X^T XW^T = 0$$

$$\Rightarrow X^T XW^T = VW X^T XW^T$$

Assuming $WX^T XW^T$ is invertible:

$$V = X^T XW^T (WX^T XW^T)^{-1}$$

$$V = X^T XW^T (W^T)^{-1} (X)^{-1} (X^T)^{-1} (W)^{-1}$$

Choosing $V = W^T$ satisfies this equation when W consists of the top k eigenvectors of $X^T X$.

We know that the covariance matrix for the dataset, can be written as:

$$\Sigma = \frac{1}{N} \sum_{n=1}^N x^{(n)} (x^{(n)})^T = \frac{1}{N} X^T X$$

$$\Sigma = U \Lambda U^T$$

where U is orthogonal and $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_d)$. The optimal weights satisfy:

$$W^* = U_k^T, \quad V^* = U_k$$

where U_k contains the top k eigenvectors.

The reconstruction becomes:

$$\hat{x} = U_k U_k^T x$$

which is the PCA reconstruction.

1. The optimal weights satisfy $\boxed{V = W^T}$.
2. The optimal W^* consists of the top k eigenvectors of Σ :

$$\boxed{W^* = U_k^T}$$